LAB 7 Switch Configuration, VLAN & Inter-VLAN Routing

Objectives:

- To configure a switch and create VLANs.
- To assign switch ports to VLANs.
- To enable communication between different VLANs using Inter-VLAN routing.
- To verify connectivity using ping.

Software Requirement:

- Cisco Packet Tracer (any recent version)
- Windows/Linux PC

Theory:

Switch

A switch is an intelligent networking device used to connect multiple devices such as computers, printers, servers, and access points within a Local Area Network (LAN). Unlike hubs, which broadcast data to all connected devices, a switch forwards data only to the intended destination device.

A switch operates primarily at the Data Link Layer (Layer 2) of the OSI model and uses MAC addresses to make forwarding decisions. When a frame arrives, the switch examines the destination MAC address and consults its MAC address table to determine the correct outgoing port.

Functions of a Switch

1. Frame forwarding – Sends data to the correct destination port.
2. MAC address learning – Builds a MAC table by observing incoming frames.
3. Filtering – Prevents unnecessary traffic on other ports.
4. Loop prevention – Uses protocols like STP to avoid network loops.
5. Segmentation – Divides collision domains, improving performance.

Advantages of Switch

- Faster communication compared to hub
- Reduced collisions
- Better bandwidth utilization
- Improved network performance
- Supports VLAN and trunking

Switch Configuration

Switch configuration is the process of setting up a switch to perform required network operations such as VLAN creation, port assignment, trunking, and security settings. Switch configuration ensures proper device identification, security, and remote management. Switches are configured using Cisco IOS CLI through console, Telnet, or SSH.

VLAN (Virtual Local Area Network)

A VLAN is a logical grouping of network devices within a switch that creates separate broadcast domains regardless of physical location. VLANs allow network administrators to segment a large LAN into smaller, manageable networks.

Devices in the same VLAN can communicate directly, while devices in different VLANs require routing through a Layer-3 device.

Why VLAN is Needed

In a flat network, broadcast traffic increases as the network grows, reducing performance and security. VLAN solves this by isolating traffic.

Types of VLAN

1. **Default VLAN** – VLAN 1 (exists by default)
2. **Data VLAN** – Used for user traffic
3. **Voice VLAN** – Used for VoIP traffic

4. **Management VLAN** – Used for switch management
5. **Native VLAN** – Untagged VLAN on trunk link

Advantages of VLAN

- **Improved security** – Isolates sensitive departments
- **Reduced broadcast traffic** – Minimizes congestion
- **Better network management** – Logical grouping of users
- **Flexibility** – Users can move without re-cabling
- **Scalability** – Supports network growth

Inter-VLAN Routing

Since VLANs create separate broadcast domains, devices in different VLANs cannot communicate directly. Inter-VLAN routing enables communication between VLANs using a router or Layer-3 switch.

It works by forwarding packets from one VLAN to another through a routing device that understands IP addresses.

Methods of Inter-VLAN Routing

1. Legacy Inter-VLAN Routing

- Uses multiple physical router interfaces
- Each interface connects to one VLAN
- Not scalable

2. Router-on-a-Stick

- Uses a single router interface
- Sub-interfaces created for each VLAN
- Requires trunk link between switch and router

- Most commonly used in labs

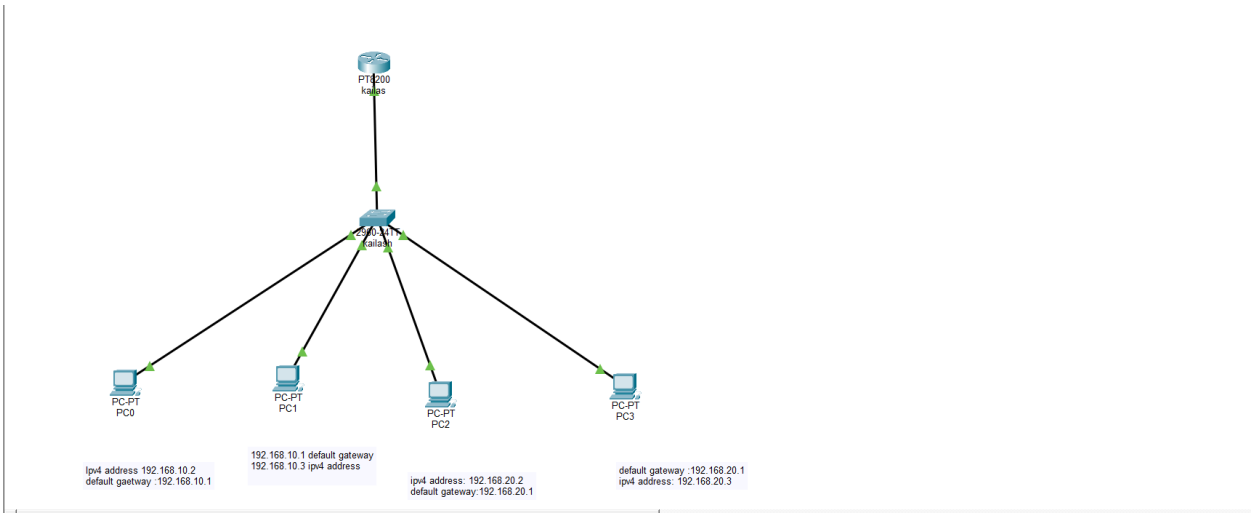
3. Layer-3 Switch Routing

- Routing performed inside the switch
- Faster and scalable
- Used in enterprise networks

Advantages of Inter-VLAN Routing

- Enables communication between departments
- Maintains VLAN isolation with connectivity
- Improves network organization
- Supports policy enforcement and security

Network Topology



The figure shows a network topology implementing VLAN with Inter-VLAN routing using the Router-on-a-Stick method. In this setup, a router is connected to a switch through a trunk link, and the switch connects to four PCs. The switch is configured with two VLANs: VLAN 10 and VLAN 20. PC0 and PC1 belong to VLAN 10 with IP addresses 192.168.10.2 and 192.168.10.3, using

192.168.10.1 as their default gateway. Similarly, PC2 and PC3 belong to VLAN 20 with IP addresses 192.168.20.2 and 192.168.20.3, using 192.168.20.1 as their default gateway. Devices within the same VLAN can communicate directly through the switch, while communication between different VLANs requires routing. The router performs this routing using sub-interfaces configured for each VLAN and acts as the default gateway for all PCs. Thus, the topology demonstrates logical network segmentation using VLANs along with successful communication between VLANs through the router.

Configuration table:

VLAN and IP Addressing Scheme

VLAN ID	VLAN Name	Network Address	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device IP Configuration

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Procedure:

1. Create network topology in Packet Tracer
2. Connect router, switch, and PCs
3. Create VLANs on switch
4. Assign switch ports to VLANs
5. Configure trunk link between switch and router
6. Configure router sub-interfaces for each VLAN
7. Assign IP addresses to PCs

8. Set default gateway on PCs
9. Verify VLAN and trunk configuration
10. Test connectivity using ping

Commands:

1.Switch Configuration Commands:

Create VLANs

Switch(config)# vlan 10

Switch(config-vlan)# name Computer

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name Electronics

Switch(config-vlan)# exit

Assign Ports to VLANs

Switch(config)# interface range fa0/1-2

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 10

Switch(config-if-range)# exit

Switch(config)# interface range fa0/3-4

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 20

Switch(config-if-range)# exit

Configure Trunk Port (Switch to Router)

Switch(config)# interface fa0/5

Switch(config-if)# switchport mode trunk

Switch(config-if)# exit

Verification

Switch# show vlan brief

```
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#switch(configure-if)#switchport mode trunk
                ^
% Invalid input detected at '^' marker.

Switch(config)#interface fa0/5
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
Switch(config)#show vlan brief
                ^
% Invalid input detected at '^' marker.

Switch(config)#
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Computer                         active    Fa0/1, Fa0/2
20   Electronics                      active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

2.Router Configuration Commands

Configure Sub-Interfaces for VLANs:

Router(config)# interface FastEthernet0/0.10

Router(config-subif)# encapsulation dot1Q 10

Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config-subif)# exit

Router(config)# interface FastEthernet0/0.20

Router(config-subif)# encapsulation dot1Q 20

Router(config-subif)# ip address 192.168.20.1 255.255.255.0

Router(config-subif)# exit

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet1/0
Router(config-if)#
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1.255.255.255.0
                                ^
% Invalid input detected at '^' marker.

Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface FastEthernet 0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

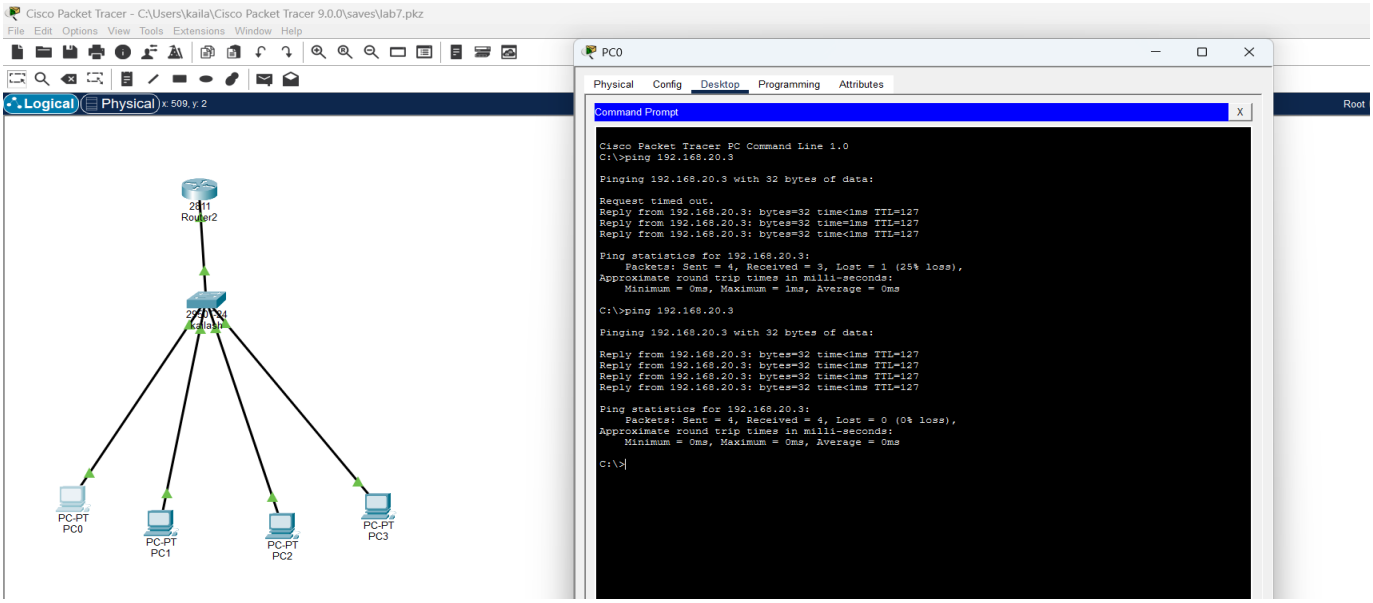
Router Verification

Router# show ip interface brief

```
Router>enable
Router# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  up                    up
FastEthernet0/0.10     192.168.10.1    YES manual up                    up
FastEthernet0/0.20     192.168.20.1    YES manual up                    up
FastEthernet1/0        unassigned      YES unset  administratively down down
Serial2/0              unassigned      YES unset  administratively down down
Serial3/0              unassigned      YES unset  administratively down down
FastEthernet4/0        unassigned      YES manual down                  down
FastEthernet5/0        unassigned      YES unset  administratively down down
Router#
```

Copy Paste

Pinging from PC0 to PC3



Discussion

This lab demonstrated the configuration of VLANs to logically divide a network into separate broadcast domains. Devices within the same VLAN communicated directly through the switch, while communication between different VLANs required routing. Inter-VLAN routing was achieved using a router connected through a trunk link, where sub-interfaces acted as default gateways for each VLAN. The experiment showed how VLANs improve security, reduce broadcast traffic, and enhance network management while still allowing controlled communication between networks.

Result

VLAN 10 and VLAN 20 were successfully created and switch ports were assigned correctly. The trunk link between the switch and router was configured, and router sub-interfaces enabled inter-VLAN routing. Devices in the same VLAN communicated directly, and successful ping responses confirmed communication between different VLANs.

Conclusion

The experiment successfully implemented VLAN configuration and Inter-VLAN routing using the Router-on-a-Stick method. The lab highlighted how VLANs segment networks for better performance and security while routing enables communication between VLANs. This technique is widely used in real-world enterprise networks for efficient traffic management.